Step 1: Connect the Router to the ISP Line

The **first device** that should be connected to the ISP line is the **router**.

Reason:

- The router acts as the **gateway** between the internet and the café's local network (LAN).
- It receives the internet connection from the ISP through the WAN (Internet) port.
- It assigns IP addresses to all connected devices using **DHCP (Dynamic Host Configuration Protocol)**.
- It provides **NAT (Network Address Translation)**, allowing multiple PCs to share a single public IP address.
- It includes a **firewall** that protects the network from unauthorized access and cyber threats.
- It manages internet traffic efficiently so that IPL streaming and online gaming work smoothly.

Step 2: Connect the Switch to the Router

After connecting the router, connect one of the router's **LAN ports** to the **8-port switch** using an Ethernet cable.

Reason:

- A router usually has only **4 LAN ports**, which are not enough for 8 gaming PCs.
- The switch increases the number of available network ports.
- It enables all computers to communicate with each other at high speed.
- A switch forwards data only to the intended device, reducing unnecessary network traffic and improving performance.

Step 3: Connect the 8 Gaming PCs

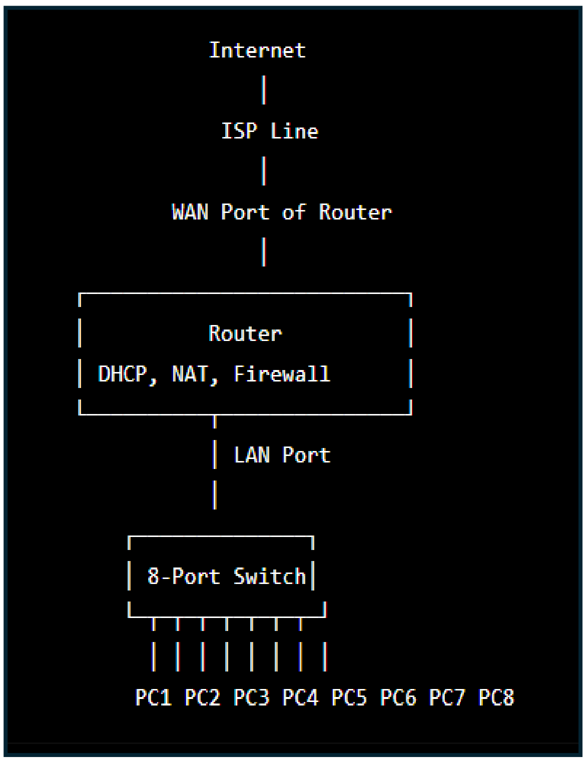
Connect each gaming PC to the switch using **Cat6 Ethernet cables**.

Each PC will automatically receive:

- An IP address
- Subnet mask
- Default gateway
- DNS server

These settings are provided by the router through DHCP.

Network Diagram



Advantages of This Network Setup

- High-speed internet access for all gaming PCs.
- Smooth IPL live streaming without buffering.
- Low-latency multiplayer gaming.

- Centralized network management through the router.
- Secure network with firewall and NAT protection.
- Easy to maintain and expand in the future.